Exercise Sheet 3

Strong and Weak Ties

5942UE Network Science

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3.A Triadic Closure

Exercise 3.A.1: Open Triads and Predicted Closures

Consider the following friendship network (all edges are **strong** ties unless stated otherwise):

Edges: A–B, A–C, A–D, B–C, D–E, E–F, D–F.

1. List all **open triads** in the network.
2. For each open triad, which missing edge is predicted to appear by **triadic closure**? Rank them by "closure pressure".
3. Which open triad, if closed, would increase the clustering coefficient of node A the most?
4. Node G joins and forms strong ties with A and D. Predict which edge forms next.

Solution:

1. Open triads: $(B, A, D), (C, A, D), (A, D, E), (A, D, F)$.

2. Ranking: B–D ≈ C–D > A–E ≈ A–F. Nodes sharing a strong-tie neighbour (A) feel the strongest pressure.

3. A's clustering: If B–D closes, A's clustering coefficient increases significantly because its neighbors B and D become connected.

4. Node G: G, A, and D form a closed triad immediately (since A–D exists). G will likely next close triads with neighbors of A (B, C) or D (E, F).

TODO: check

Exercise 3.A.2: Computing Clustering Coefficients

For the graph: A–B, A–C, A–D, B–C, D–E, E–F, D–F:

1. Compute the **local clustering coefficient** C_v for every node.
2. Compute the **average clustering coefficient** $\bar{C}$.
3. Interpret the result: what does it tell us about the network?
4. Visualise the graph with nodes coloured by C_v.

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Solution:

1. Local coefficients:

- $C_B, C_C, C_E, C_F = 1$ (their neighbours are all connected).
- $C_A = 1/3$ (neighbours B, C, D, only B, C connected).
- $C_D = 1/3$ (neighbours A, E, F, only E, F connected).

2. Average: $\bar{C} = (1 + 1 + 1 + 1 + 1/3 + 1/3)/6 = 14/18 = 7/9 \approx 0.78$.

3. Interpretation: $\bar{C} \approx 0.78$ is high, indicating a "cliquey" social structure where friends of friends are likely to be friends.

TODO: check

3.B Bridges and Brokers

Exercise 3.B.1: Finding Bridges and Structural Holes

Three cliques: $1, 2, 3$, $4, 5, 6$, and $7, 8, 9$. Bridges: 3–4 and 6–7.

1. Is 3–4 a bridge? Is 6–7 a bridge?
2. Is 3–4 a **local bridge**?
3. Node 5 in Clique 2 connects to Cliques 1 and 3. Does it sit in a **structural hole**?
4. If node 3 forms a direct edge with node 7, what happens to the hole and the bridges?

Solution:

- **1. Bridges:** Yes. Removing either disconnects the graph into separate cliques (components).
- **2. Local Bridge:** Yes. Nodes 3 and 4 have no common neighbours besides each other.
- **3. Structural Hole:** Clique 1 and Clique 3 have no direct connection. Node 5 (via 4 and 6) can broker information between these disjoint worlds, gaining an information advantage.
- **4. Edge 3–7:** The structural hole closes. Edge $6–7$ is no longer a bridge because an alternative path (via 3–7) now exists.

TODO: check

Exercise 3.B.2: Identifying Bridges with NetworkX

Build the three-clique graph in NetworkX. Identify bridges and rank edges by edge betweenness.

Solution:

```
import networkx as nx

G = nx.Graph()
G.add_edges_from([(1,2),(1,3),(2,3),        # Clique 1
                  (4,5),(4,6),(5,6),         # Clique 2
                  (7,8),(7,9),(8,9),         # Clique 3
                  (3,4),(6,7)])              # Bridges

# 1. Bridges
bridges = list(nx.bridges(G))
print("Bridges:", bridges)   # [(3,4), (6,7)]

# 2. Edge betweenness
ebc = nx.edge_betweenness_centrality(G)
sorted_ebc = sorted(ebc.items(), key=lambda x: -x[1])
for edge, bc in sorted_ebc[:2]:
    print(f"  {edge}: {bc:.4f}")
```

Interpretation:

- **High Betweenness:** Bridges (3,4) and (6,7) carry all traffic between cliques, giving them maximum edge betweenness.
- **Vulnerability:** Removing these edges fragments the network faster than removing internal clique edges, demonstrating the importance of "weak" bridging ties for global connectivity.

3.C Weak Ties Theorem

Exercise 3.C.1: Verifying the Weak-Ties Theorem

Weak-Ties Theorem: Under Strong Triadic Closure (STC), every local bridge is a weak tie.

Check STC and local bridges for:

- **A:** Strong $1–2, 1–3$. Weak $2–4$. No $2–3$.
- **B:** Strong $1–2, 1–3, 2–3$. Weak $1–4$.
- **C:** Strong $1–2, 2–3, 1–3, 3–4, 2–4$.

Solution:

- **A: STC Violated** (1 tied to 2 and 3, but no 2–3 edge). Theorem premise doesn't hold.
- **B: STC Satisfied**. Edge 1–4 is a local bridge and is **weak**. Theorem holds.
- **C: STC Violated** (2 tied to 1 and 4, but no 1–4 edge). Theorem doesn't apply.

Exercise 3.C.2: Granovetter's Job Study

1. Why are rarely-seen acquaintances more useful for job finding than close friends?
2. Apply the weak-ties theorem to redundant vs. novel information.
3. Is "unfriending all close friends" good advice based on the theorem?
4. How does LinkedIn fit this framework?

Solution:

- **1. Novelty:** Friends share your world and information. Acquaintances bridge to different social circles, providing **non-redundant** job leads.
- **2. Structure:** Strong ties cluster in dense groups (redundancy). Weak ties are bridges to distant clusters (novelty).
- **3. Support vs Reach:** No. Strong ties provide trust and support; weak ties provide reach. You need both.
- **4. LinkedIn:** Formalises weak ties with low neighborhood overlap, enabling people to scale their "reach" into structural holes.

3.D Evidence and Overlap

Exercise 3.D.1: Neighbourhood Overlap Calculation

$$O(u, v) = \frac{|\mathcal{N}(u) \cap \mathcal{N}(v)|}{|\mathcal{N}(u) \cup \mathcal{N}(v)|}$$

(excluding u and v from $\mathcal{N}$)

For edges: A-B, A-C, A-D, B-C, B-E, C-F:

1. Compute $O(A, B)$, $O(A, C)$, and $O(B, E)$.
2. Rank them. Which is most like a local bridge?

Solution:

- $O(A, B)$ = $C / C, D, E$ = **1/3**
- $O(A, C)$ = $B / B, D, F$ = **1/3**
- $O(B, E)$ = $\emptyset / A, C$ = **0**

Ranking: $O(B, E)$ has zero overlap, making it a perfect **local bridge**. Information entering via E is likely to be completely novel to the A, B, C cluster.

Exercise 3.D.2: Neighbourhood Overlap in NetworkX

Compute average neighborhood overlap within and across factions in the karate club.

Solution:

```python
import networkx as nx
import numpy as np

G = nx.karate_club_graph()

def neighborhood_overlap(G, u, v):
    nu = set(G.neighbors(u)) - {v}
    nv = set(G.neighbors(v)) - {u}
    if not (nu or nv): return 0.0
    return len(nu & nv) / len(nu | nv)

# Compare within-faction vs across-faction
within = [neighborhood_overlap(G, u, v) for u, v in G.edges()
          if G.nodes[u]["club"] == G.nodes[v]["club"]]
across = [neighborhood_overlap(G, u, v) for u, v in G.edges()
          if G.nodes[u]["club"] != G.nodes[v]["club"]]

print(f"Avg Overlap Within: {np.mean(within):.3f}")
print(f"Avg Overlap Across: {np.mean(across):.3f}")
```

Interpretation:

- **Cross-faction weakness:** Cross-faction edges have significantly lower overlap, confirming they behave as structural bridges between communities.
- **Glue vs redundancy:** Within-faction edges have high overlap, providing structural redundancy. Low-overlap edges are the "glue" holding separate social worlds together.

Exercise 3.D.3: Practical Lab — Weak Ties in the Karate Club

Use `nx.karate_club_graph()` to connect the lecture concepts to measurable graph quantities.

1. Compute the **local clustering coefficient** C_v for every node.
2. Compute the **neighbourhood overlap** $O(u, v)$ for every edge.
3. Classify edges as:
   - **weak / bridge-like** if $O(u, v) < 0.10$
   - **medium** if $0.10 \leq O(u, v) < 0.30$
   - **strong / embedded** if $O(u, v) \geq 0.30$
4. Visualise the graph:
   - node colour = clustering coefficient
   - edge colour and width = overlap-based tie strength
5. List the five lowest-overlap edges. Are they mostly within one faction or between factions?

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Solution:

```
import networkx as nx
import matplotlib.pyplot as plt

G = nx.karate_club_graph()

def neighbourhood_overlap(G, u, v):
    nu = set(G.neighbors(u)) - {v}
    nv = set(G.neighbors(v)) - {u}
    union = nu | nv
    if not union:
        return 0.0
    return len(nu & nv) / len(union)

# 1. Node-level embeddedness
clustering = nx.clustering(G)

# 2. Edge-level embeddedness
overlap = {
    (u, v): neighbourhood_overlap(G, u, v)
```

Visualisation:

```
pos = nx.spring_layout(G, seed=7)

edge_colours = []
edge_widths = []
for edge in G.edges():
    o = overlap[edge]
    if o < 0.10:
        edge_colours.append("#d95f02")   # weak / bridge-like
        edge_widths.append(1.0)
    elif o < 0.30:
        edge_colours.append("#7570b3")   # medium
        edge_widths.append(2.0)
    else:
        edge_colours.append("#1b9e77")   # strong / embedded
        edge_widths.append(4.0)

node_colours = [clustering[n] for n in G.nodes()]

plt.figure(figsize=(9, 7))
```

Interpretation:

- **Low-overlap edges** are the best structural candidates for weak ties. They have few or no shared neighbours, so they connect otherwise separate local neighbourhoods.
- **High-overlap edges** sit inside dense parts of the graph. They are structurally redundant and are reasonable candidates for strong, embedded ties.
- **Faction-crossing edges** usually have lower overlap than within-faction edges, but overlap is a structural proxy, not ground-truth psychology. Some within-faction ties can also be bridge-like if they connect two sparse neighbourhoods.

3.E Tie Strength in Practice

Exercise 3.E.1: Tie Strengths in Practice

1. Facebook (symmetric) vs. Twitter (asymmetric): structural implications for overlap?
2. Why do maintained relationships grow slowly while total count grows quickly?
3. How should you design a recommender for **information diversity**?

Solution:

- **1. Symmetry:** Facebook ties are reciprocal, leading to higher local clustering and overlap. Twitter links are closer to structural bridges with near-zero overlap.
- **2. Cognitive limits:** Maintained ties are bounded by the Dunbar number and social attention. Passive links are structural weak ties with high novelty but low investment.
- **3. Diversity:** Recommend content from **low-overlap connections**. These structural bridges provide access to disjoint social bubbles and non-redundant information.

Exercise 3.E.2: Simulating Tie Strength and Overlap

Assign synthetic strength values to karate-club edges (within-faction strong, across-faction weak) and compute the correlation with neighborhood overlap.

Solution:

```
import networkx as nx
import numpy as np

G = nx.karate_club_graph()

# Assign synthetic strength: within-faction = strong (0.8), cross = weak (0.2)
strengths, overlaps = [], []
for u, v in G.edges():
    s = 0.8 if G.nodes[u]["club"] == G.nodes[v]["club"] else 0.2
    strengths.append(s)
    overlaps.append(neighborhood_overlap(G, u, v))  # defined in 3.D.2

# Correlation
r = np.corrcoef(strengths, overlaps)[0, 1]
print(f"Pearson correlation: {r:.3f}")  # Typically > 0.6
```

Interpretation:

- **Empirical support:** The strong positive correlation confirms that tie strength and neighborhood overlap are linked: close-knit groups share friends, while bridging ties remain socially isolated.
- **Structural exceptions:** High-strength ties with low overlap may indicate bridging relationships that involve significant social effort (e.g., cross-community collaborations).